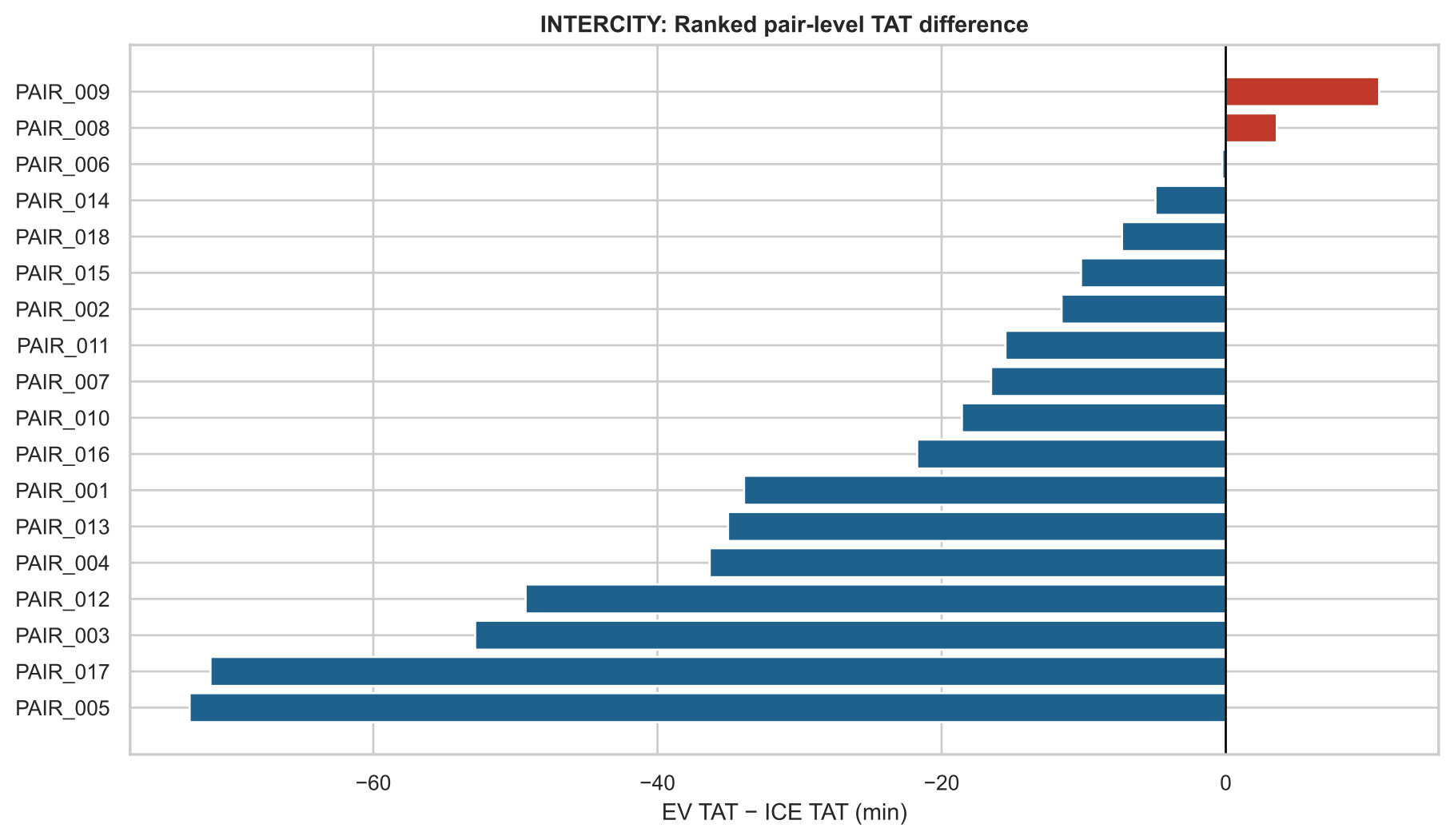
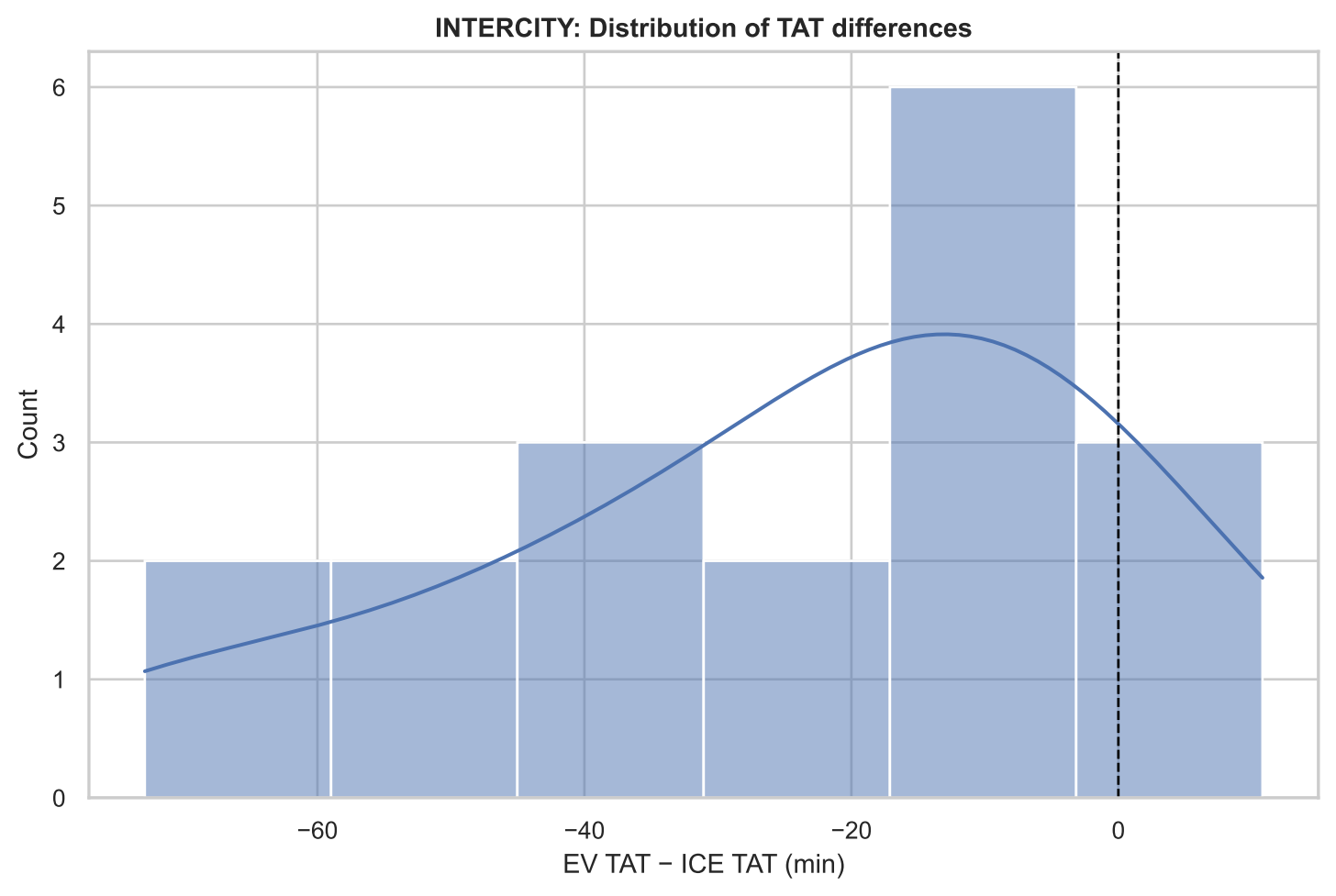
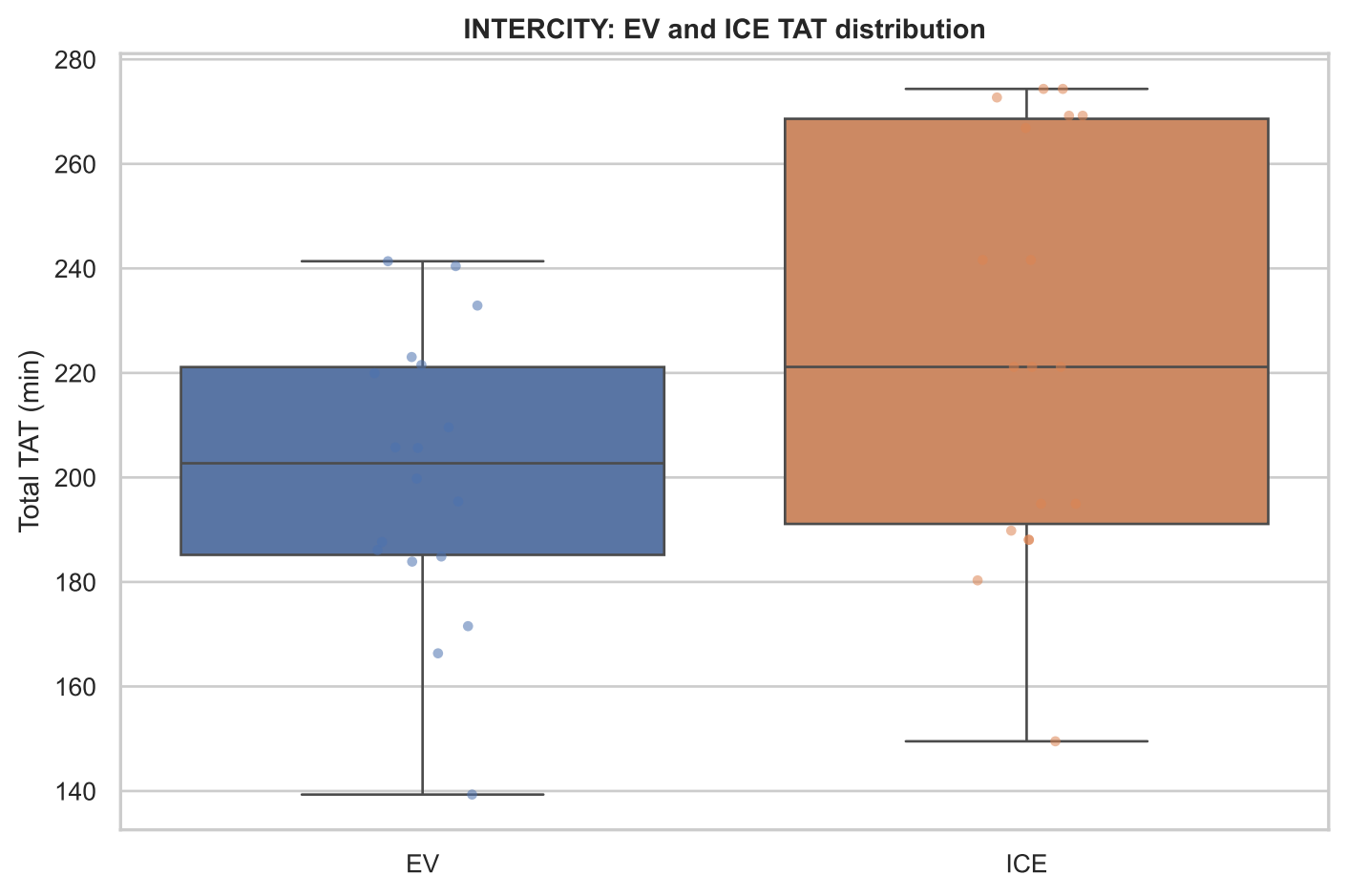
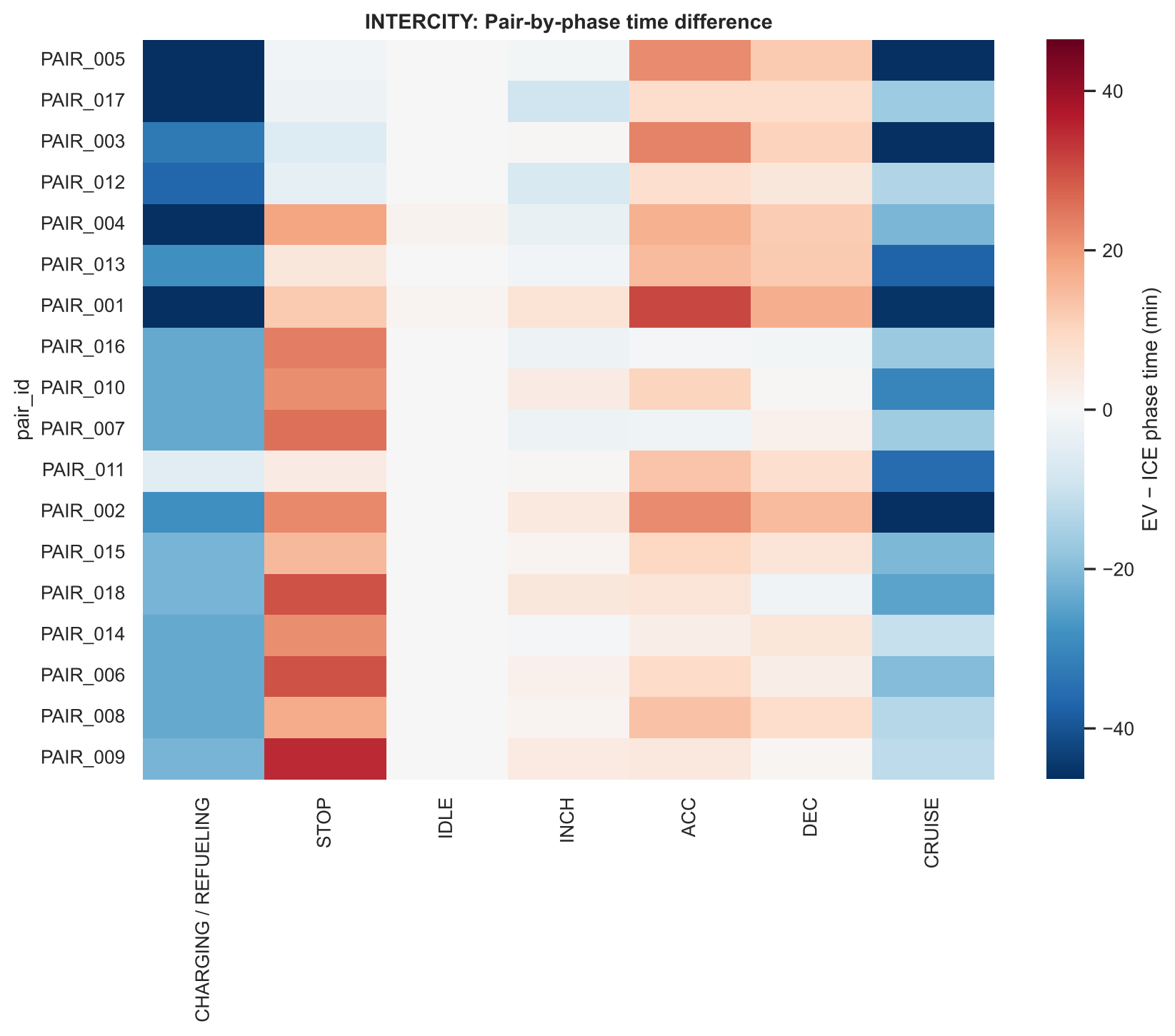
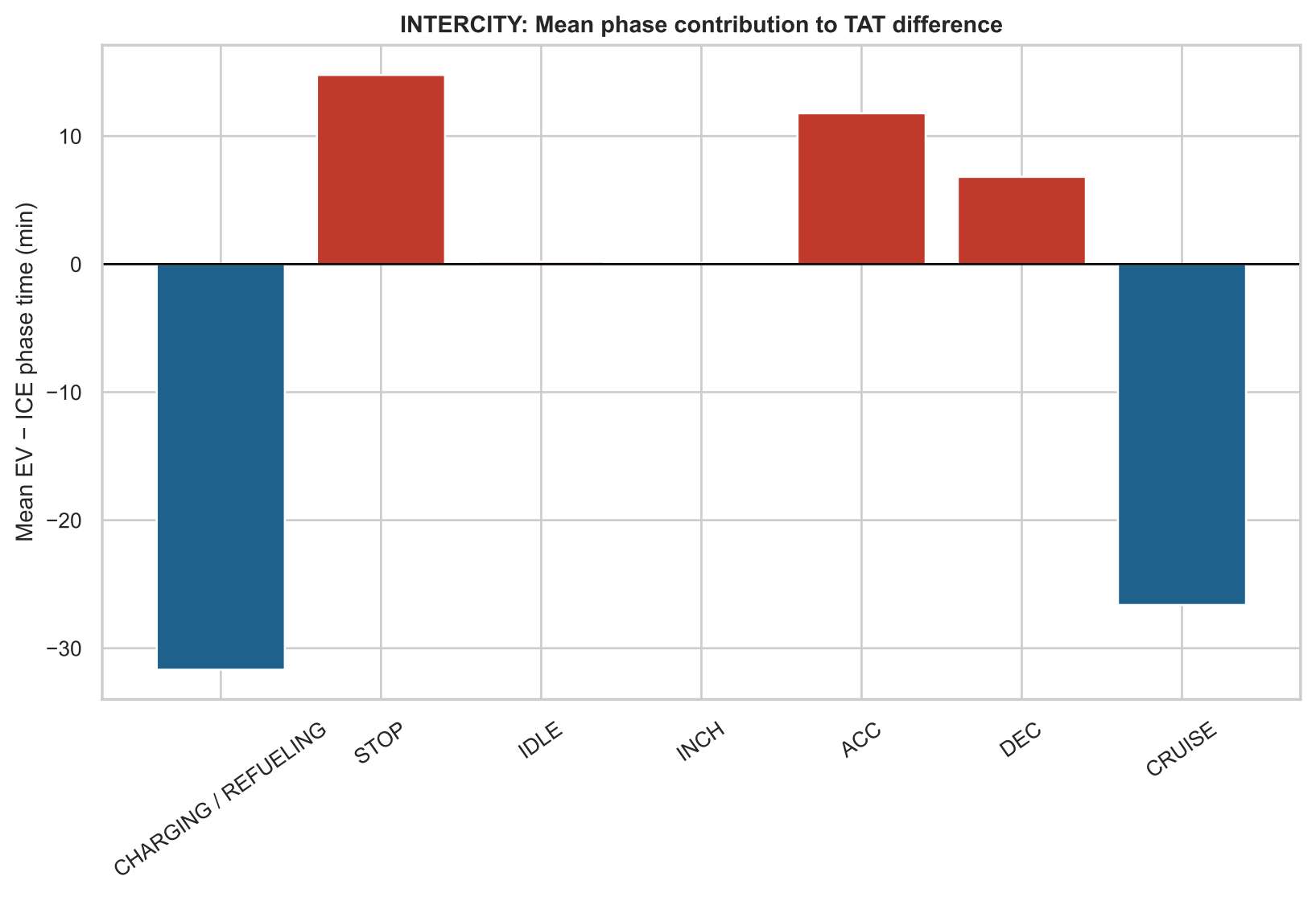
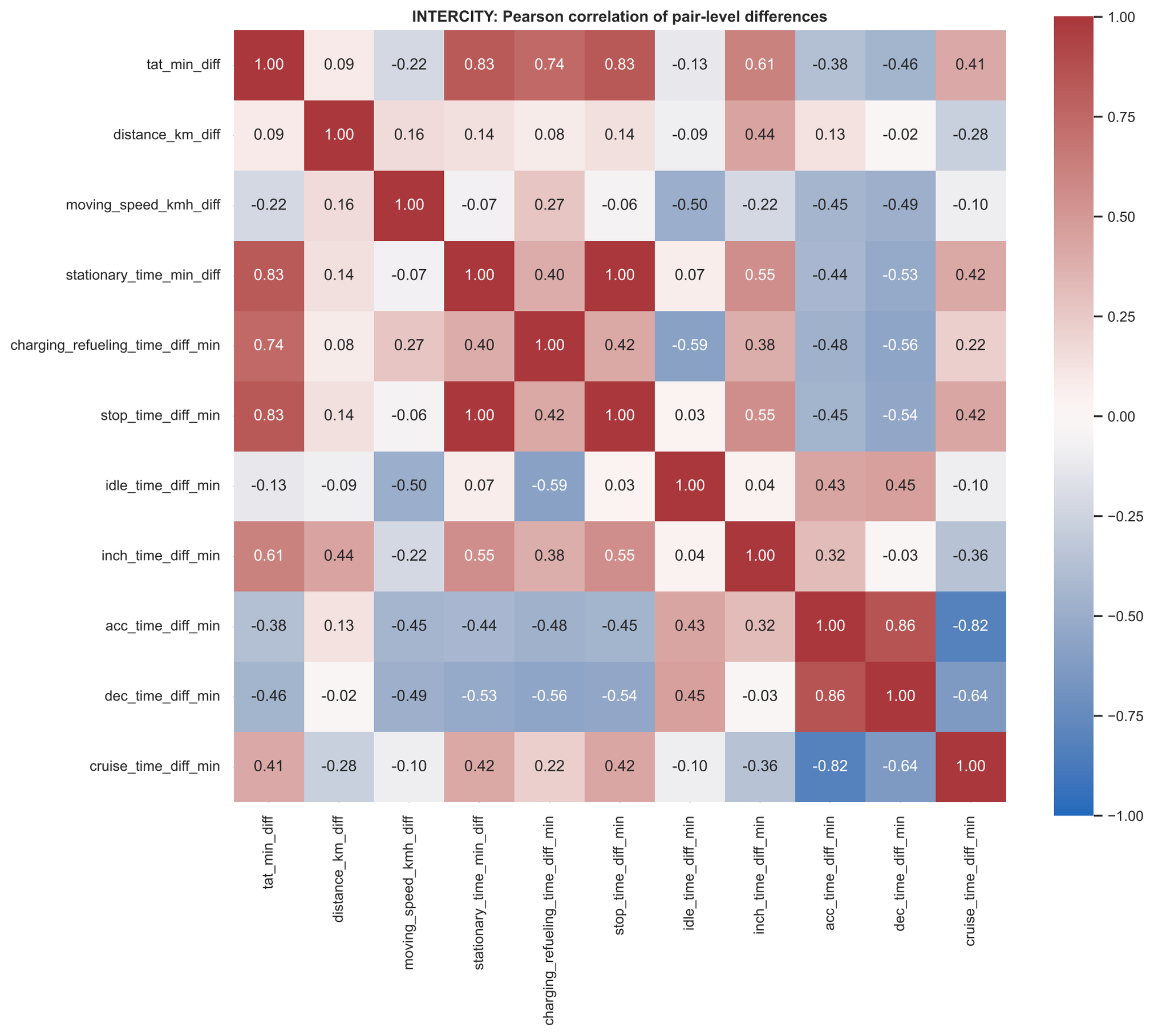

INTERCITY: Ranked pair-level TAT difference
PAIR_009
PAIR_008
PAIR_006
PAIR_014
PAIR_018
PAIR_015
PAIR_002
PAIR_011
PAIR_007
PAIR_010
PAIR_016
PAIR_001
PAIR_013
PAIR_004
PAIR_012
PAIR_003
PAIR_017
PAIR_005
−60
−40
−20
0
EV TAT − ICE TAT (min)

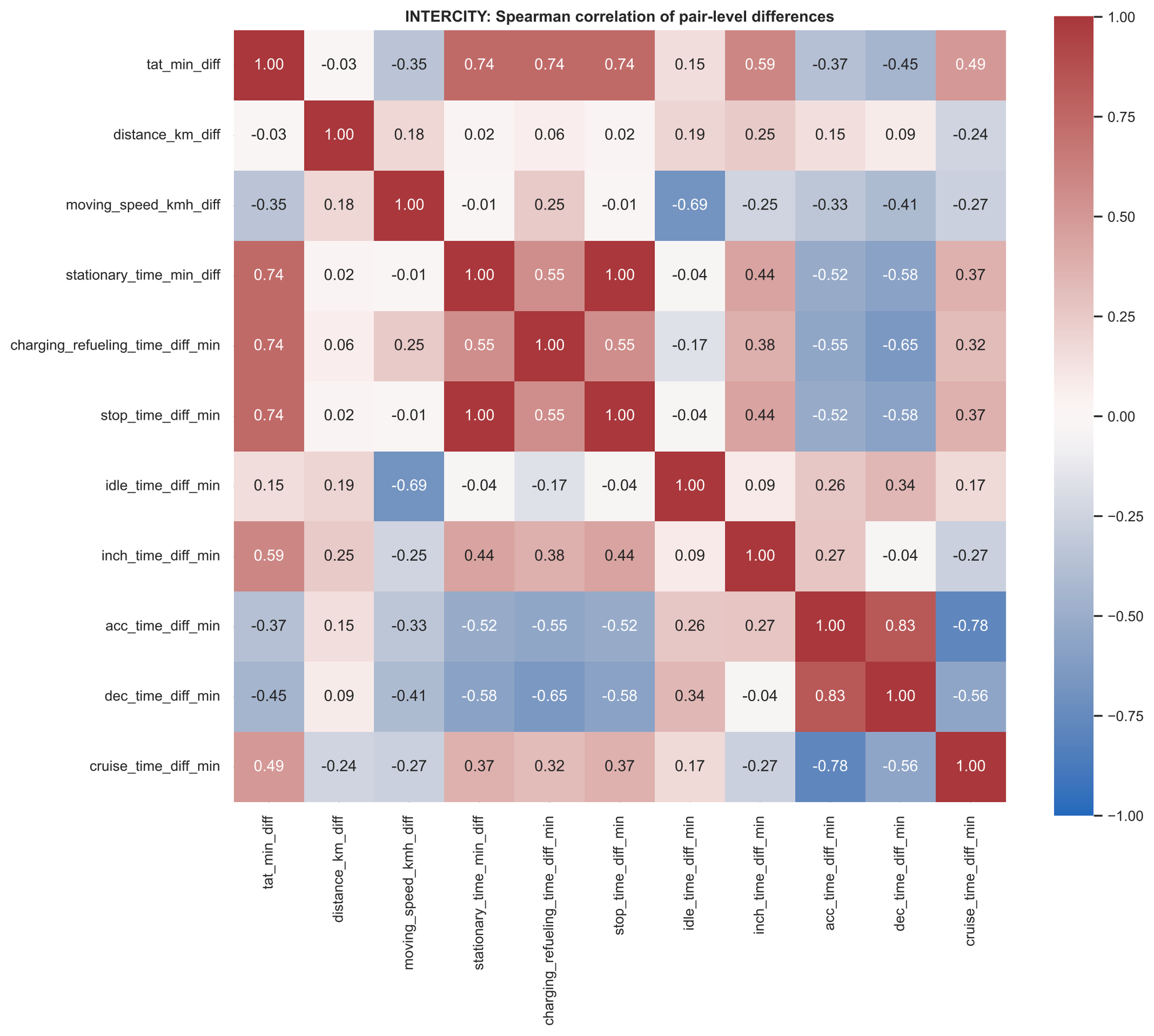

INTERCITY: Distribution of TAT differences
Count
0
1
2
3
4
5
6
−60
−40
−20
0
EV TAT − ICE TAT (min)

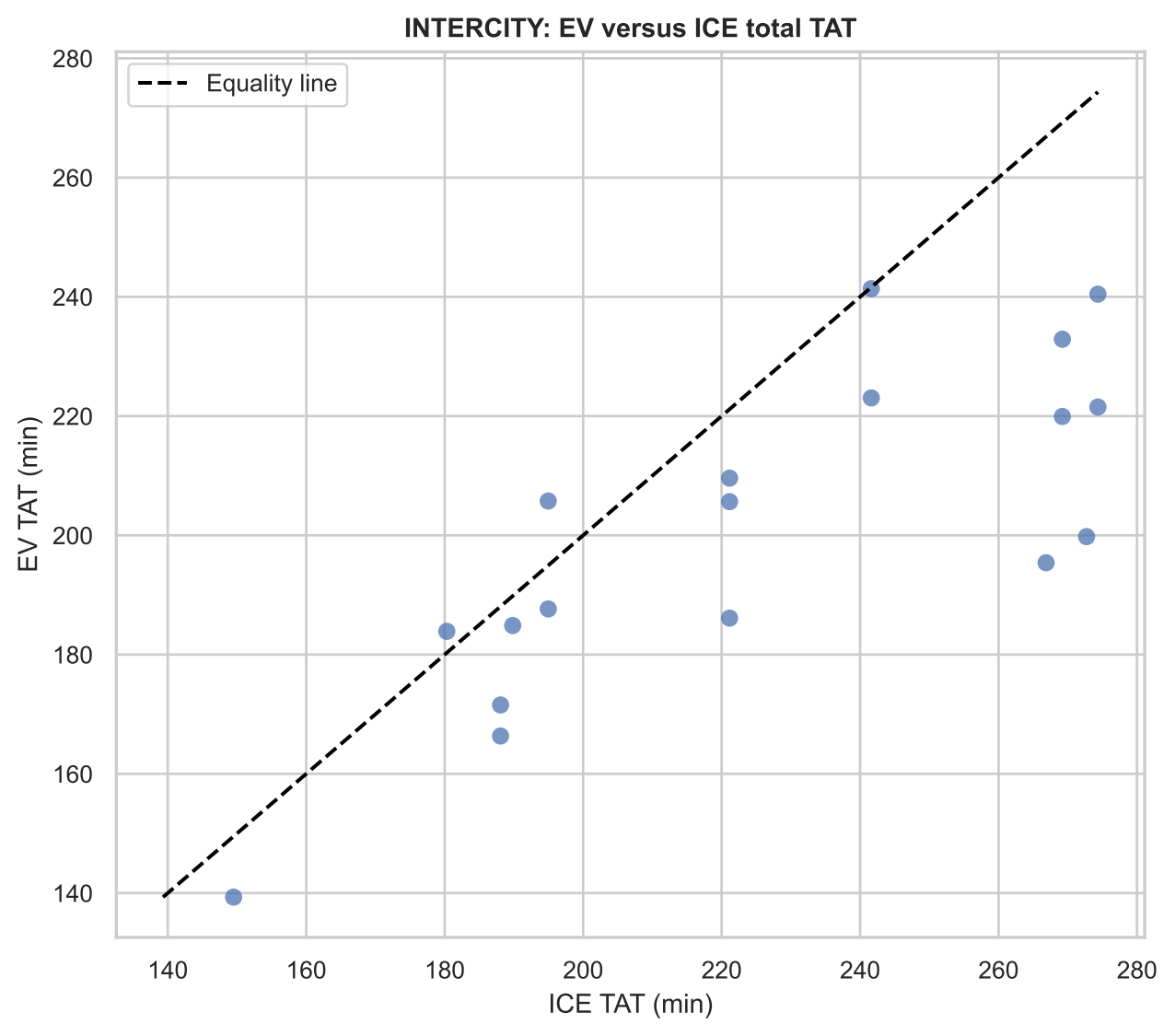

INTERCITY: EV and ICE TAT distribution
Total TAT (min)
280
260
240
220
200
180
160
140
EV
ICE

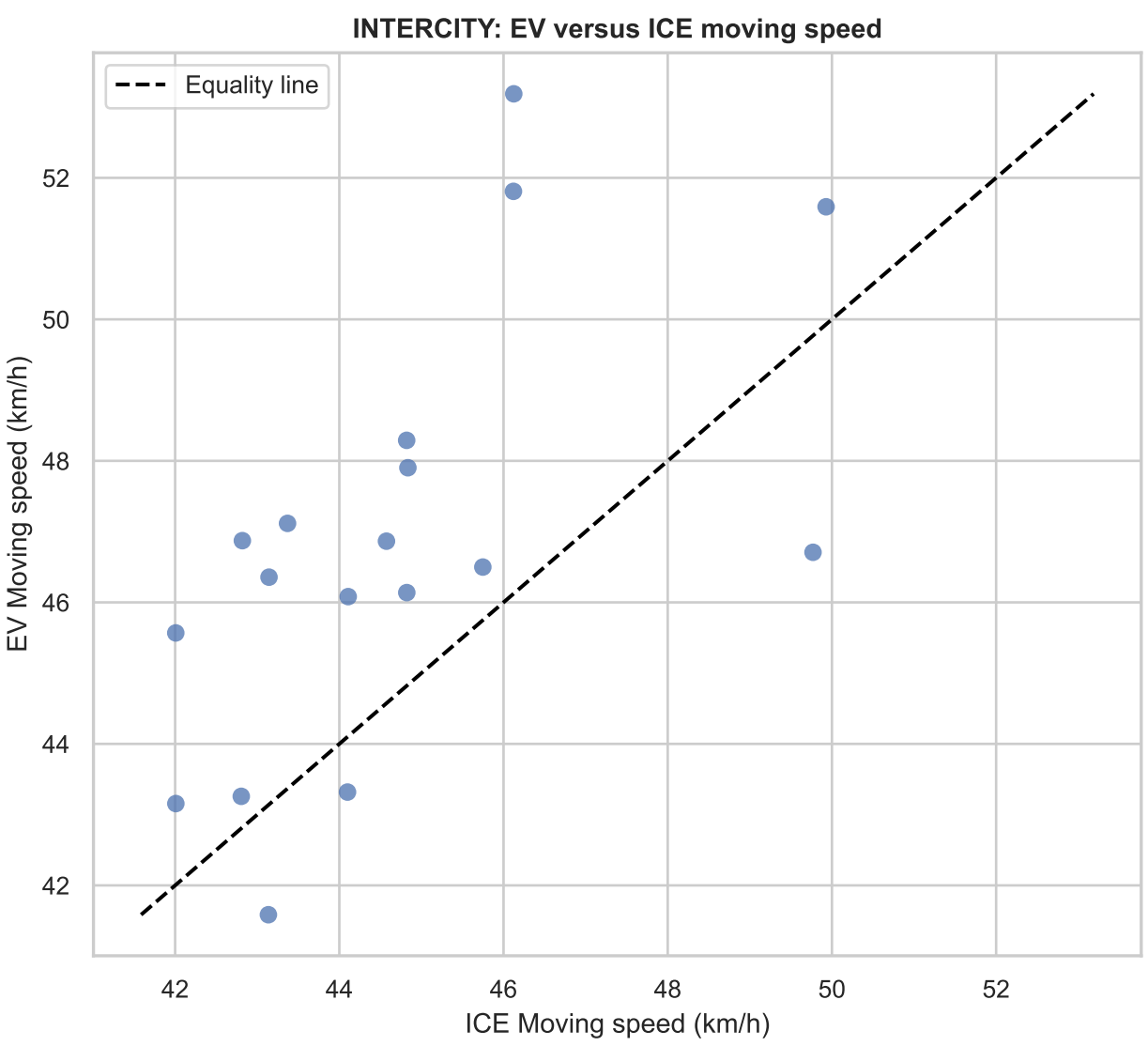

INTERCITY: Pair-by-phase time difference
pair_id
PAIR_005
PAIR_017
PAIR_003
PAIR_012
PAIR_004
PAIR_013
PAIR_001
PAIR_016
PAIR_010
PAIR_007
PAIR_011
PAIR_002
PAIR_015
PAIR_018
PAIR_014
PAIR_006
PAIR_008
PAIR_009
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE
EV − ICE phase time (min)
40
20
0
−20
−40

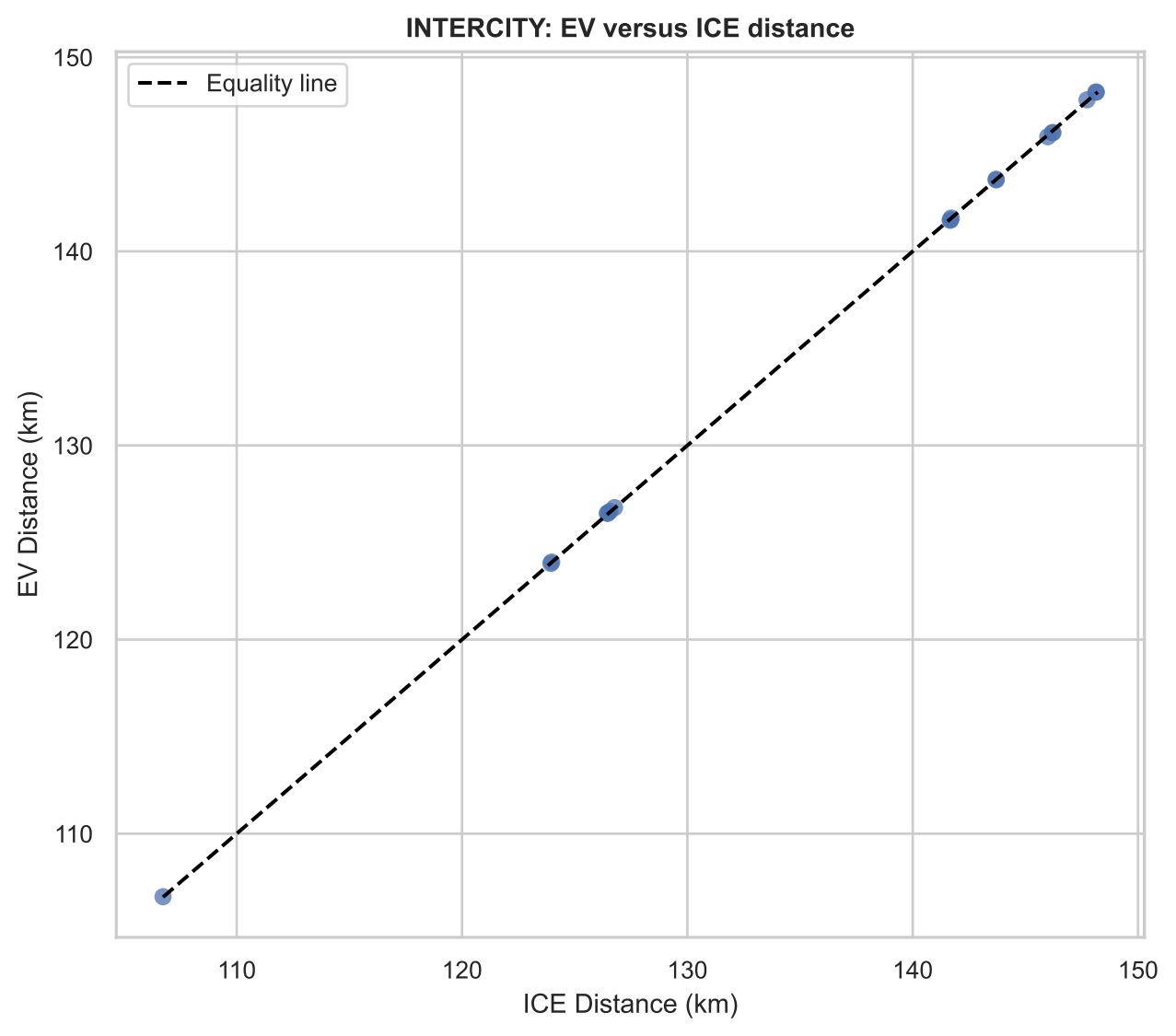

INTERCITY: Mean phase contribution to TAT difference
Mean EV − ICE phase time (min)
10
0
−10
−20
−30
CHARGING / REFUELING
STOP
IDLE
INCH
ACC
DEC
CRUISE

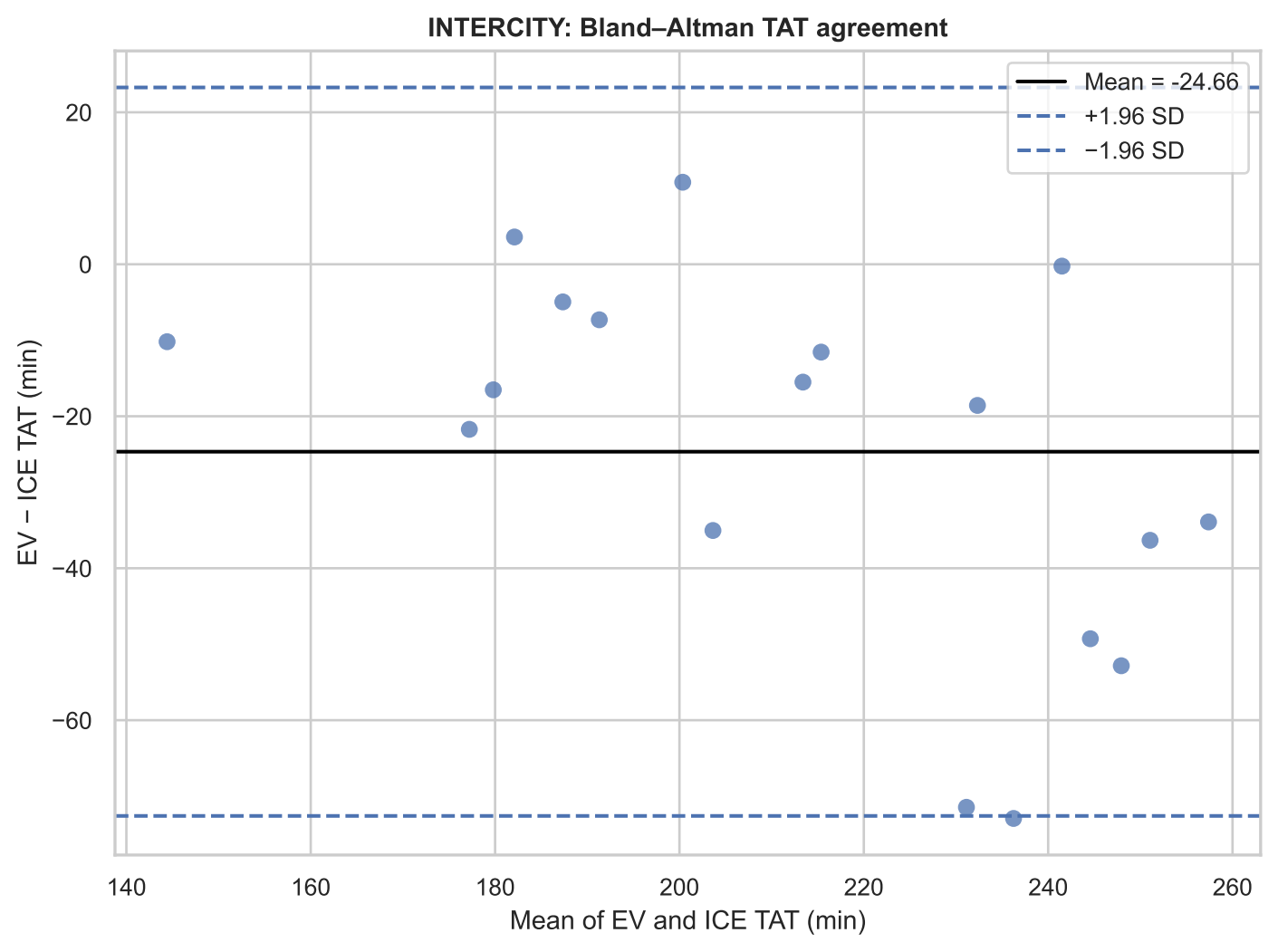

INTERCITY: Pearson correlation of pair-level differences
tat_min_diff 1.00 0.09 -0.22 0.83 0.74 0.83 -0.13 0.61 -0.38 -0.46 0.41
distance_km_diff 0.09 1.00 0.16 0.14 0.08 0.14 -0.09 0.44 0.13 -0.02 -0.28
moving_speed_kmh_diff -0.22 0.16 1.00 -0.07 0.27 -0.06 -0.50 -0.22 -0.45 -0.49 -0.10
stationary_time_min_diff 0.83 0.14 -0.07 1.00 0.40 1.00 0.07 0.55 -0.44 -0.53 0.42
charging_refueling_time_diff_min 0.74 0.08 0.27 0.40 1.00 0.42 -0.59 0.38 -0.48 -0.56 0.22
stop_time_diff_min 0.83 0.14 -0.06 1.00 0.42 1.00 0.03 0.55 -0.45 -0.54 0.42
idle_time_diff_min -0.13 -0.09 -0.50 0.07 -0.59 0.03 1.00 0.04 0.43 0.45 -0.10
inch_time_diff_min 0.61 0.44 -0.22 0.55 0.38 0.55 0.04 1.00 0.32 -0.03 -0.36
acc_time_diff_min -0.38 0.13 -0.45 -0.44 -0.48 -0.45 0.43 0.32 1.00 0.86 -0.82
dec_time_diff_min -0.46 -0.02 -0.49 -0.53 -0.56 -0.54 0.45 -0.03 0.86 1.00 -0.64
cruise_time_diff_min 0.41 -0.28 -0.10 0.42 0.22 0.42 -0.10 -0.36 -0.82 -0.64 1.00
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

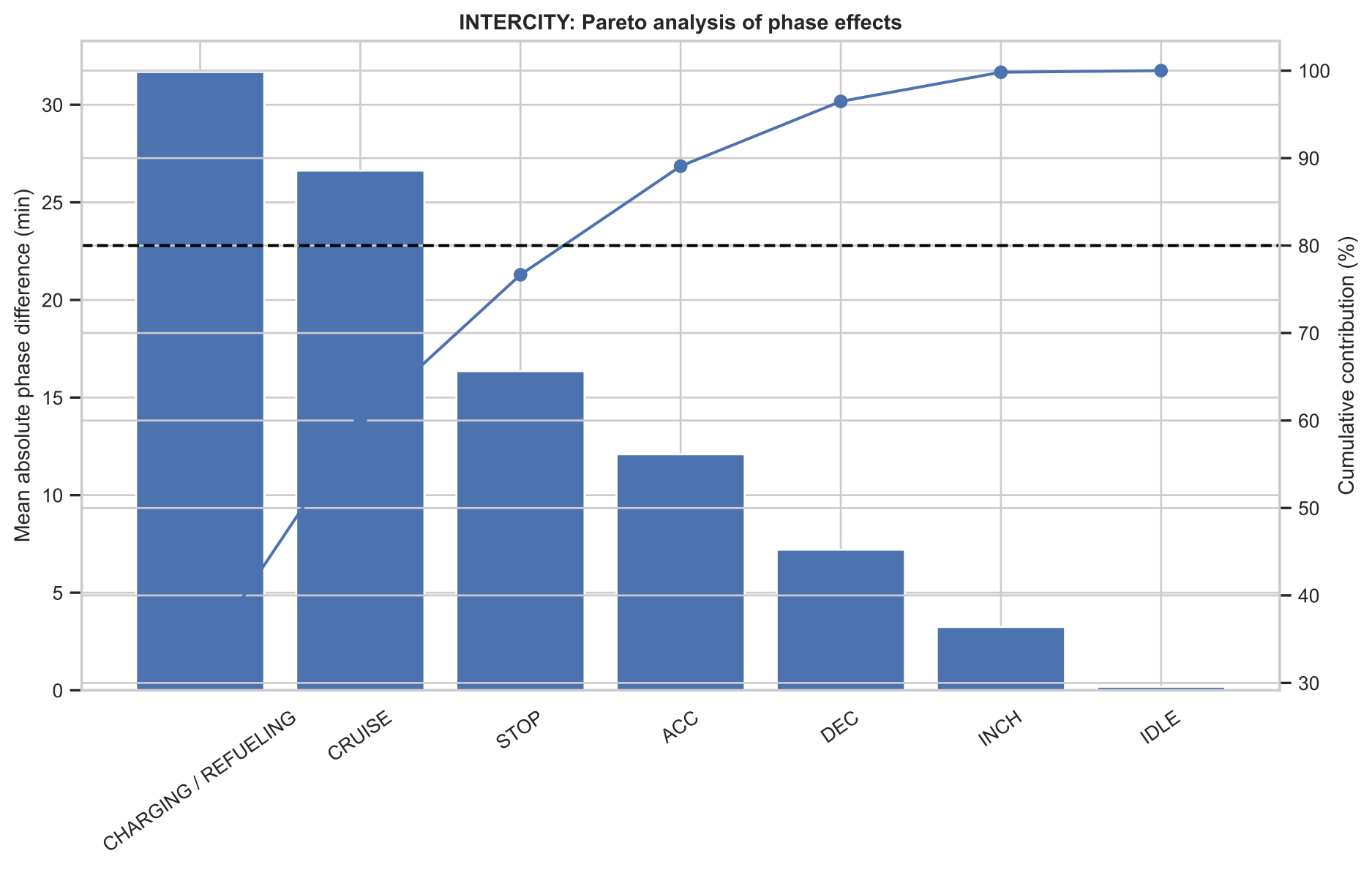

INTERCITY: Spearman correlation of pair-level differences
tat_min_diff 1.00 -0.03 -0.35 0.74 0.74 0.74 0.15 0.59 -0.37 -0.45 0.49
distance_km_diff -0.03 1.00 0.18 0.02 0.06 0.02 0.19 0.25 0.15 0.09 -0.24
moving_speed_kmh_diff -0.35 0.18 1.00 -0.01 0.25 -0.01 -0.69 -0.25 -0.33 -0.41 -0.27
stationary_time_min_diff 0.74 0.02 -0.01 1.00 0.55 1.00 -0.04 0.44 -0.52 -0.58 0.37
charging_refueling_time_diff_min 0.74 0.06 0.25 0.55 1.00 0.55 -0.17 0.38 -0.55 -0.65 0.32
stop_time_diff_min 0.74 0.02 -0.01 1.00 0.55 1.00 -0.04 0.44 -0.52 -0.58 0.37
idle_time_diff_min 0.15 0.19 -0.69 -0.04 -0.17 -0.04 1.00 0.09 0.26 0.34 0.17
inch_time_diff_min 0.59 0.25 -0.25 0.44 0.38 0.44 0.09 1.00 0.27 -0.04 -0.27
acc_time_diff_min -0.37 0.15 -0.33 -0.52 -0.55 -0.52 0.26 0.27 1.00 0.83 -0.78
dec_time_diff_min -0.45 0.09 -0.41 -0.58 -0.65 -0.58 0.34 -0.04 0.83 1.00 -0.56
cruise_time_diff_min 0.49 -0.24 -0.27 0.37 0.32 0.37 0.17 -0.27 -0.78 -0.56 1.00
tat_min_diff
distance_km_diff
moving_speed_kmh_diff
stationary_time_min_diff
charging_refueling_time_diff_min
stop_time_diff_min
idle_time_diff_min
inch_time_diff_min
acc_time_diff_min
dec_time_diff_min
cruise_time_diff_min
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00

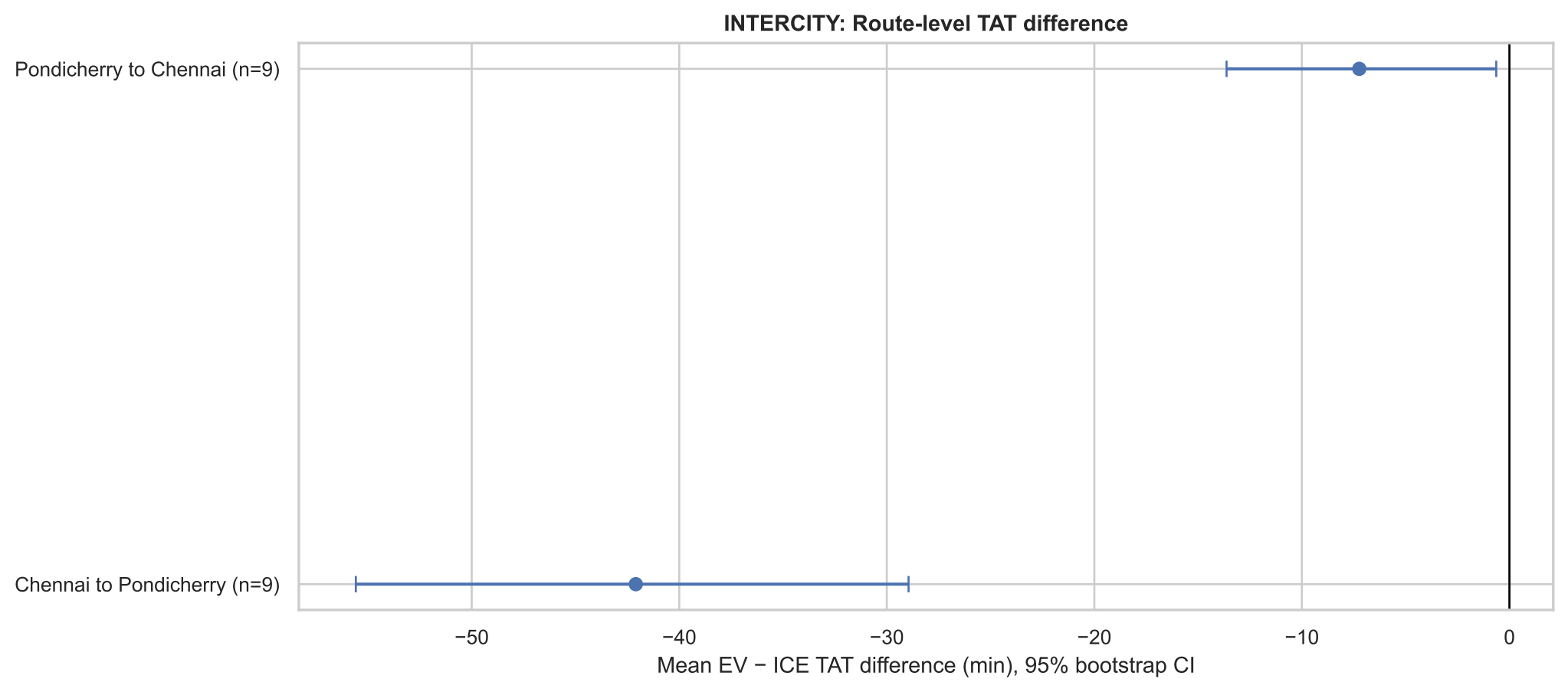

INTERCITY: EV versus ICE total TAT
Equality line
EV TAT (min)
ICE TAT (min)
280
260
240
220
200
180
160
140
140
160
180
200
220
240
260
280

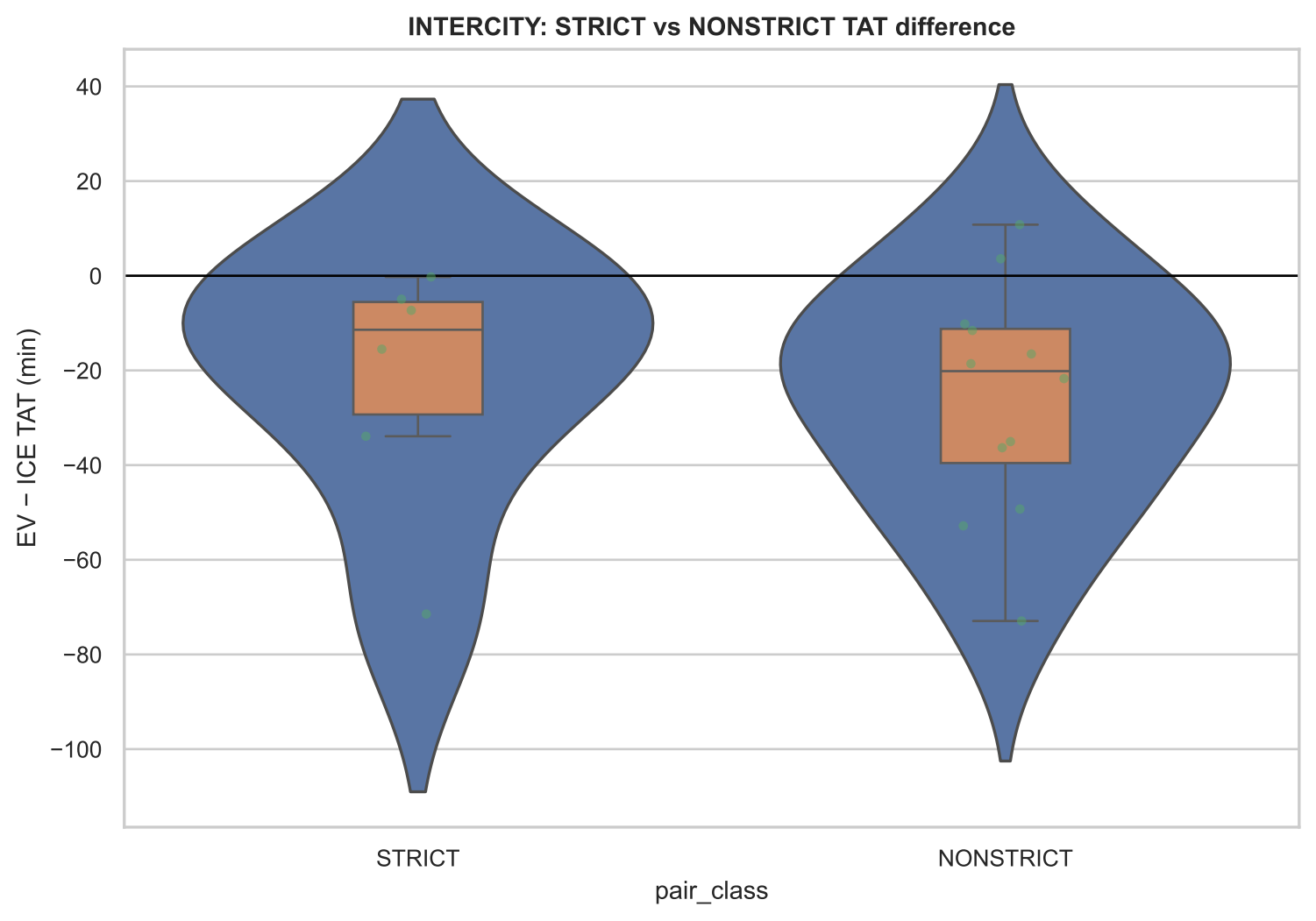

INTERCITY: EV versus ICE moving speed
Equality line
EV Moving speed (km/h)
ICE Moving speed (km/h)
42
44
46
48
50
52

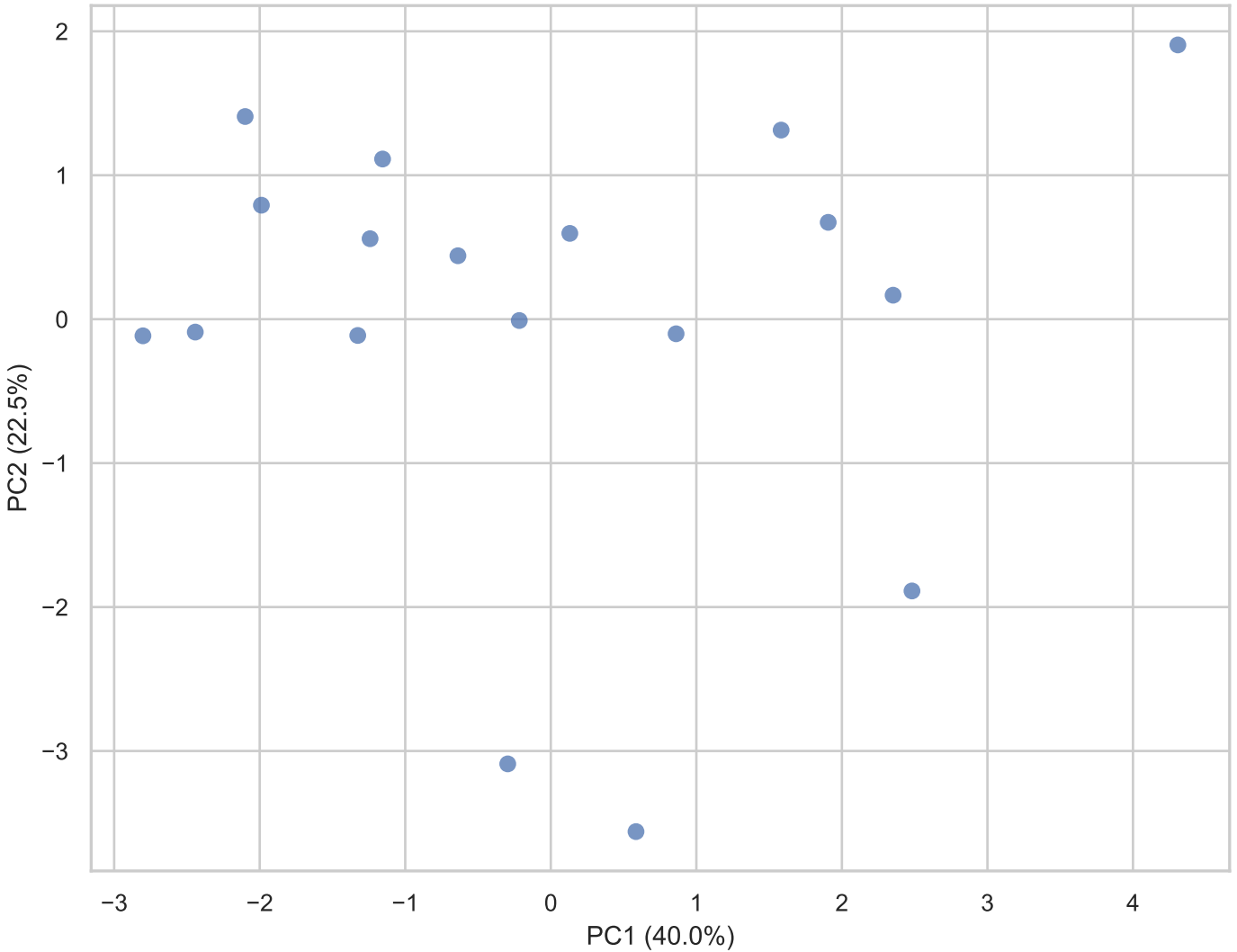

INTERCITY: EV versus ICE distance
Equality line
EV Distance (km)
ICE Distance (km)
150
140
130
120
110
110
120
130
140
150

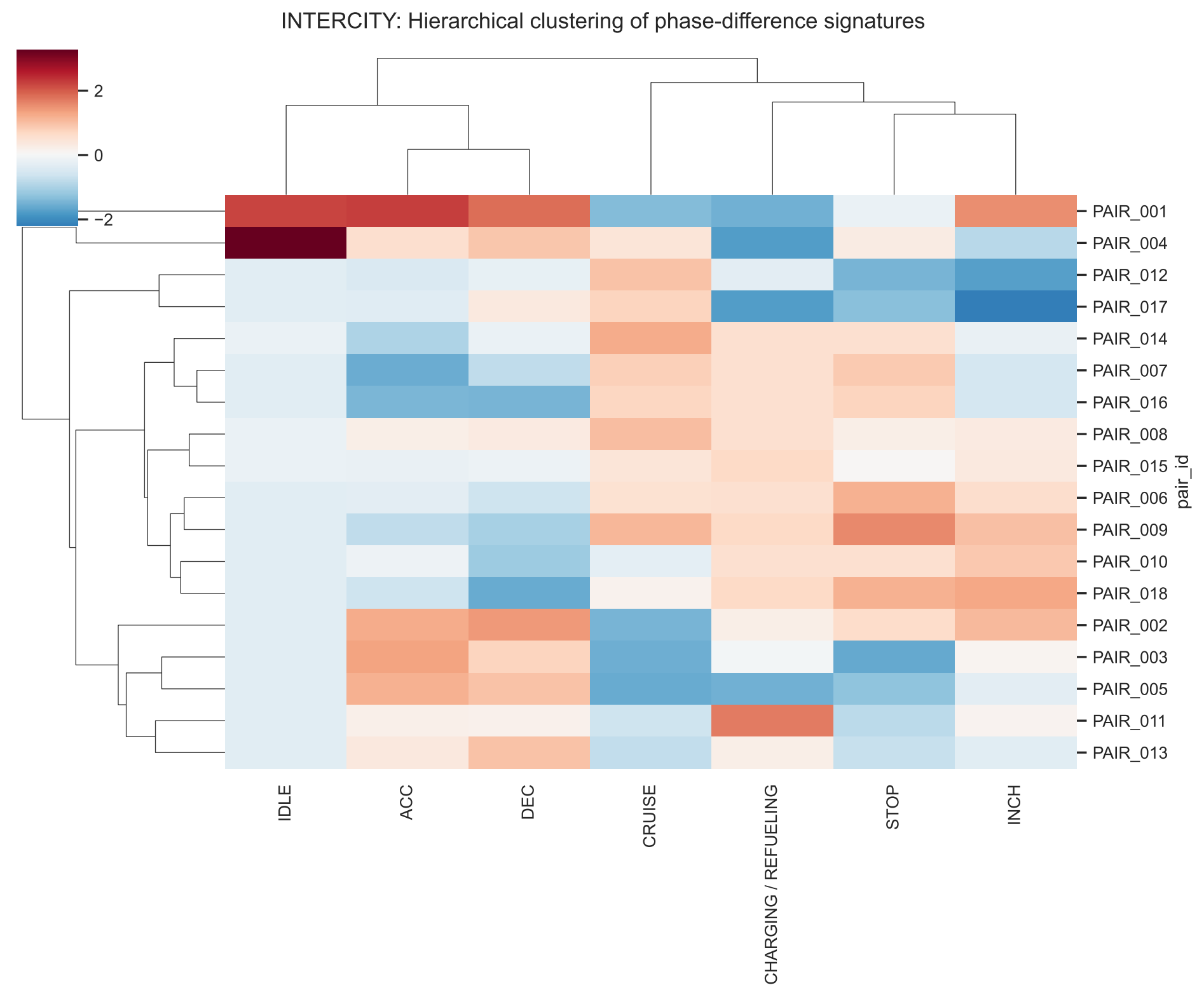

INTERCITY: Bland–Altman TAT agreement
Mean = -24.66
+1.96 SD
−1.96 SD
20
0
−20
−40
−60
EV − ICE TAT (min)
140
160
180
200
220
240
260
Mean of EV and ICE TAT (min)


INTERCITY: Pareto analysis of phase effects
Mean absolute phase difference (min)
Cumulative contribution (%)
0
5
10
15
20
25
30
30
40
50
60
70
80
90
100
CHARGING / REFUELING
CRUISE
STOP
ACC
DEC
INCH
IDLE


INTERCITY: Route-level TAT difference
Pondicherry to Chennai (n=9)
Chennai to Pondicherry (n=9)
−50
−40
−30
−20
−10
0
Mean EV − ICE TAT difference (min), 95% bootstrap CI


INTERCITY: STRICT vs NONSTRICT TAT difference
40
20
0
−20
−40
−60
−80
−100
EV – ICE TAT (min)
STRICT
NONSTRICT
pair_class


INTERCITY: PCA of EV−ICE operational differences
PC2 (22.5%)
PC1 (40.0%)
2
1
0
−1
−2
−3
−3
−2
−1
0
1
2
3
4


INTERCITY: Hierarchical clustering of phase-difference signatures
2
0
−2
PAIR_001
PAIR_004
PAIR_012
PAIR_017
PAIR_014
PAIR_007
PAIR_016
PAIR_008
PAIR_015
PAIR_006
PAIR_009
PAIR_010
PAIR_018
PAIR_002
PAIR_003
PAIR_005
PAIR_011
PAIR_013
pair_id
IDLE
ACC
DEC
CRUISE
CHARGING / REFUELING
STOP
INCH